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**1. Entity-Relationship Diagram (ERD)**

**Refined ERD Components**

**Entities and Attributes**

1. **Volunteer**
    - **VolunteerID** (Primary Key)
    - FirstName
    - LastName
    - ContactInfo (Phone, Email, Address)
    - LicenseNumber
    - Availability (Days, Times)
    - Skills (Haircut, Beard Trim, Styling)
    - CertificationStatus
    - JoinDate
    - Status (Active, Inactive)
2. **Service**
    - **ServiceID** (Primary Key)
    - ServiceType (Haircut, Beard Trim, Styling, etc.)
    - Date
    - Time
    - Duration
    - MobileUnitID (Foreign Key to MobileUnit)

- VolunteerID (Foreign Key to Volunteer)
- ClientID (Foreign Key to Client)
- Feedback (Optional)

3. **Donation**
    - **DonationID** (Primary Key)
    - DonorID (Foreign Key to Donor)
    - Amount
    - Date
    - PaymentMethod (Credit Card, PayPal, Bank Transfer, etc.)
    - Campaign (if applicable)
    - ReceiptSent (Boolean)
4. **Donor**
    - **DonorID** (Primary Key)
    - DonorName
    - ContactInfo (Phone, Email, Address)
    - DonorType (Individual, Corporate)
    - DonationHistory (Foreign Key to Donation)
    - PreferredContactMethod
    - MembershipStatus (Recurring, One-time)
5. **Event**
    - **EventID** (Primary Key)
    - EventName
    - Location
    - Date
    - Description
    - PartnerID (Foreign Key to Partnership)

- FundsRaised
- Participants (Volunteers, Clients)

6. **MobileUnit**
    - **UnitID** (Primary Key)
    - UnitName
    - CurrentLocation
    - AvailableServices (List of ServiceTypes)
    - MaintenanceStatus
    - AssignedVolunteers (Foreign Key to Volunteer)
    - Capacity
7. **Partnership**
    - **PartnerID** (Primary Key)
    - PartnerName
    - PartnerType (Shelter, Business, Healthcare, etc.)
    - ContactInfo
    - PartnershipStartDate
    - PartnershipEndDate
    - ResourcesProvided (Supplies, Funding, Space)
    - Location (Shelter, Community Center, Hospital, etc.)
    - ServiceHistory (Foreign Key to Service)
    - ClientID
    - ClientFirstName
    - ClientLastName
    - ClientDOB
    - ClientGender
    - ClientContactInfo

- ClientRegistrationDate
- ClientStatus (Active, Inactive)

8. **Appointment**
    - **AppointmentID** (Primary Key)
    - ClientID (Foreign Key to Client)
    - VolunteerID (Foreign Key to Volunteer)
    - MobileUnitID (Foreign Key to MobileUnit)
    - ServiceID (Foreign Key to Service)
    - ScheduledDate
    - ScheduledTime
    - Status (Scheduled, Completed, Canceled)
9. **Feedback**
    - **FeedbackID** (Primary Key)
    - ServiceID (Foreign Key to Service)
    - ClientID (Foreign Key to Client)
    - VolunteerID (Foreign Key to Volunteer)
    - Rating (1-5)
    - Comments
    - SubmissionDate

**Enhanced Relationships**

- **Volunteer** ↔ **Service**
    - One-to-Many: A volunteer can provide multiple services.
- **Partnership** ↔ **Service**
    - One-to-Many: A partnership can receive/schedule multiple services.
- **Service** ↔ **MobileUnit**
    - Many-to-One: Multiple services can be assigned to a single mobile unit.

- **Donation ↔ Donor**
    - Many-to-One: A donor can make multiple donations.
- **Event ↔ Partnership**
    - Many-to-One: An event can have one partner, but a partner can sponsor multiple events.
- **Appointment** connects **Partnership**, **Volunteer**, **Service**, and **MobileUnit**
    - Many-to-One: Each appointment links to one client, one volunteer, one service, and one mobile unit.
- **Feedback** connects to **Service**, **Partnership**, and **Volunteer**
    - One-to-One: Each service can have one feedback entry from a client.

# 2. Data Flow Diagrams (DFD)

## Refined DFD Components

### Context Level DFD

- **External Entities:**
    - Volunteers
    - Clients
    - Donors
    - Partners
    - Administrators (Internal)
- **Central Process:**
    - Cuts of Compassion System
- **Data Flows:**
    - **Volunteers** register, update availability, receive schedules.
    - **Clients** request services, provide feedback.
    - **Donors** submit donations, receive receipts.
    - **Partners** collaborate on events, provide resources.
    - **Administrators** manage data, generate reports.

### Level 1 DFDs

Let's break down the **Cuts of Compassion System** into major subprocesses.

#### Process 1.0: Volunteer Management

- **Inputs:**
    - Volunteer Registration Data
    - Availability Updates
- **Outputs:**

    - Volunteer Profiles
    - Schedules
    - Notifications
- **Data Stores:**
    - Volunteer Database
    - Schedule Database
- **Subprocesses:**
    - Register Volunteer
    - Update Availability
    - Assign Services

#### Process 2.0: Client Service Management

- **Inputs:**
    - Service Requests
    - Client Information
- **Outputs:**
    - Scheduled Appointments
    - Service Records
- **Data Stores:**
    - Client Database
    - Appointment Database
- **Subprocesses:**
    - Register Client
    - Book Service
    - Track Service History

#### Process 3.0: Donation Management

- **Inputs:**
    - Donation Information
    - Donor Details
- **Outputs:**
    - Donation Records
    - Receipts
- **Data Stores:**
    - Donation Database
    - Donor Database
- **Subprocesses:**
    - Process Donation
    - Generate Receipt
    - Update Donor Profile

#### Process 4.0: Event Management

- **Inputs:**
    - Event Details
    - Partnership Information
- **Outputs:**
    - Event Schedules
    - Participation Records

- **Data Stores:**
  - Event Database
  - Partnership Database
- **Subprocesses:**
  - Create Event
  - Manage Partnerships
  - Track Event Outcomes

#### Process 5.0: Reporting and Analytics

- **Inputs:**
  - Data from Volunteer, Client, Donation, and Event Databases
- **Outputs:**
  - Impact Reports
  - Financial Reports
  - Operational Dashboards
- **Data Stores:**
  - All Databases
- **Subprocesses:**
  - Generate Reports
  - Analyze Data
  - Share Insights with Stakeholders

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## Detailed Level 2 DFD for Service Booking (Process 2.0)

### Process 2.1: Register Client

- **Inputs:**
  - Client Personal Information
  - Contact Details
  - Initial Service Needs
- **Outputs:**
  - Client Profile Created
  - Welcome Notification
- **Data Stores:**
  - Client Database

### Process 2.2: Book Service Appointment

- **Inputs:**
  - Client Service Request
  - Volunteer Availability
  - Mobile Unit Availability
- **Outputs:**
  - Appointment Confirmation
  - Schedule Update
- **Data Stores:**
  - Appointment Database
  - Volunteer Database

    - MobileUnit Database

**Process 2.3: Assign Volunteer and Mobile Unit**

- **Inputs:**
    - Scheduled Appointment Details
    - Volunteer Schedules
    - Mobile Unit Locations
- **Outputs:**
    - Assigned Volunteer and Unit
    - Notification to Volunteer
- **Data Stores:**
    - Volunteer Database
    - MobileUnit Database

**Process 2.4: Conduct Service**

- **Inputs:**
    - Appointment Details
    - Client and Volunteer Information
- **Outputs:**
    - Service Record
    - Feedback Request
- **Data Stores:**
    - Service Database
    - Feedback Database

**3. Wireframe Diagrams**

**Refined Wireframe Components**

Let's detail each web form with layout suggestions, sections, and validation requirements.

**1. Volunteer Registration Form**

**Layout:**

- **Header:** "Join Our Team of Volunteers"
- **Sections:**
    - **Personal Information**
        - First Name (Text Input, Required)
        - Last Name (Text Input, Required)
        - Email Address (Email Input, Required, Validation: Format)
        - Phone Number (Text Input, Required, Validation: Numeric)

    - **Professional Details**
        - License Number (Text Input, Required)
        - Certification Status (Dropdown: Certified, In Training, Not Certified)
        - Skills (Checkboxes: Haircut, Beard Trim, Styling, Other)
    - **Availability**
        - Days Available (Checkboxes: Monday to Sunday)
        - Preferred Time Slots (Dropdown: Morning, Afternoon, Evening)
    - **Additional Information**
        - Motivation for Volunteering (Textarea, Optional)
        - Upload Certification (File Upload, Optional)
- **Buttons:**
    - Submit (Primary Button)
    - Reset (Secondary Button)
- **Footer:** "Thank you for your interest in volunteering with Cuts of Compassion!"

**Validations:**

- All required fields must be filled.
- Email must be in proper format.
- Phone number must contain only digits.
- License number must match expected format.

**Confirmation Page:**

- Message: "Thank you for registering as a volunteer! We will review your application and get back to you shortly."

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**2. Service Request Form (for Clients)**

**Layout:**

- **Header:** "Request a Complimentary Service"

- **Sections:**
    - **Client Information**
        - First Name (Text Input, Required)
        - Last Name (Text Input, Required)
        - Age (Number Input, Required)
        - Gender (Dropdown: Male, Female, Non-binary, Prefer not to say)
    - **Contact Information**
        - Phone Number (Text Input, Optional)
        - Email Address (Email Input, Optional)
    - **Service Details**
        - Service Type (Dropdown: Haircut, Beard Trim, Styling)
        - Preferred Date (Date Picker, Required)
        - Preferred Time Slot (Dropdown: Morning, Afternoon, Evening)
        - Location (Dropdown: Shelter Name, Community Center, Hospital, Other)
    - **Additional Requests**
        - Special Instructions (Textarea, Optional)
- **Buttons:**
    - Submit (Primary Button)
    - Reset (Secondary Button)
- **Footer:** "We’re here to help you look and feel your best."

**Validations:**

- Required fields must be completed.
- Date cannot be in the past.
- If "Other" is selected for Location, display a text input for specifying.

**Confirmation Page:**

- Message: "Your service request has been submitted! You will receive a confirmation once your appointment is scheduled."

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**3. Donation Form**

**Layout:**

- **Header:** "Support Our Mission with a Donation"
- **Sections:**
    - **Donor Information**
        - First Name (Text Input, Required)
        - Last Name (Text Input, Required)
        - Email Address (Email Input, Required)
        - Phone Number (Text Input, Optional)
    - **Donation Details**
        - Donation Amount (Number Input, Required, Validation: Minimum $1)
        - Donation Type (Radio Buttons: One-time, Recurring)
        - Payment Method (Dropdown: Credit Card, PayPal, Bank Transfer)
        - Campaign (Dropdown: General Fund, Mobile Unit Support, Event Sponsorship, Other)
    - **Payment Information**
        - Card Number (Text Input, Required if Credit Card)
        - Expiration Date (MM/YY, Required if Credit Card)
        - CVV (Text Input, Required if Credit Card)
        - PayPal Email (Email Input, Required if PayPal)
        - Bank Account Details (Required if Bank Transfer)
    - **Additional Information**
        - Message to the Organization (Textarea, Optional)
- **Buttons:**

    - Donate Now (Primary Button)
    - Reset (Secondary Button)
- **Footer:** "Thank you for your generosity and support!"

**Validations:**

- All required fields must be completed.
- Payment details must match the selected payment method.
- Donation amount must be a positive number.

**Confirmation Page:**

- Message: "Thank you for your donation! A receipt has been sent to your email."

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**4. Event Creation Form (for Partners)**

**Layout:**

- **Header:** "Partner with Us for Community Events"
- **Sections:**
    - **Event Information**
        - Event Name (Text Input, Required)
        - Event Description (Textarea, Required)
        - Event Date (Date Picker, Required)
        - Event Time (Time Picker, Required)
        - Event Location (Text Input, Required)
    - **Partnership Details**
        - Partner Organization Name (Text Input, Required)
        - Partner Contact Person (Text Input, Required)
        - Contact Email (Email Input, Required)
        - Contact Phone (Text Input, Optional)
    - **Resources Provided**

        - Resources Offered (Checkboxes: Financial Support, Supplies, Venue, Volunteers, Other)
        - Description of Resources (Textarea, Optional)
    - **Participation Details**
        - Number of Volunteers Needed (Number Input, Optional)
        - Expected Number of Clients (Number Input, Optional)
- **Buttons:**
    - Submit Event (Primary Button)
    - Reset (Secondary Button)
- **Footer:** "Thank you for partnering with Cuts of Compassion to make a difference!"

**Validations:**

- Required fields must be filled.
- Event date must be a future date.
- Contact email must be in proper format.

**Confirmation Page:**

- Message: "Your event has been submitted! Our team will review the details and get back to you shortly."

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**5. Admin Dashboard Wireframes**

**Layout:**

- **Sidebar Navigation:**
    - Dashboard Home
    - Volunteers
    - Clients
    - Services
    - Donations
    - Events

    - Reports
    - Settings
- **Main Content Area:**
    - **Dashboard Home:**
        - Overview Metrics (Number of Volunteers, Clients Served, Donations Received, Upcoming Events)
        - Recent Activity Feed (Latest Service Bookings, Recent Donations, New Volunteers)
        - Quick Actions (Add Volunteer, Schedule Service, Create Event)
    - **Volunteers Page:**
        - List of Volunteers (Search, Filter by Status, Export Data)
        - Volunteer Profiles (Detailed View, Edit/Delete Options)
    - **Clients Page:**
        - List of Clients (Search, Filter by Location, Export Data)
        - Client Profiles (Service History, Feedback, Edit/Delete Options)
    - **Services Page:**
        - Upcoming Services (Calendar View, Assign Volunteers)
        - Service History (Filter by Date, Volunteer, Mobile Unit)
    - **Donations Page:**
        - Donation Records (Search, Filter by Donor, Export Data)
        - Donor Profiles (Donation History, Contact Information)
    - **Events Page:**
        - Upcoming Events (Calendar View, Partner Information)
        - Event Details (Participants, Funds Raised)
    - **Reports Page:**
        - Generate Reports (Impact, Financial, Operational)
        - View and Download Reports

    - **Settings Page:**
        - Manage Users
        - System Configurations
        - Data Backup Options

**Features:**

- **Responsive Design:** Ensure the dashboard is accessible on various devices.
- **User Roles:** Different access levels for admins, managers, and staff.
- **Real-time Updates:** Live data syncing for accurate reporting.

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****6. Additional Wireframes**

**User Login/Register Pages:**

- **Login Page:**
    - Fields: Email, Password
    - Buttons: Login, Forgot Password, Register
    - Links: "Are you a volunteer?" / "Are you a partner?"
- **Register Page:**
    - Options to register as Volunteer, Donor, or Partner
    - Relevant fields based on selection
    - Buttons: Register, Reset

**Profile Pages:**

- **Volunteer Profile:**
    - Personal Information
    - Availability
    - Assigned Services
    - Feedback Received
- **Client Profile:**

    - Personal Information
    - Service History
    - Feedback Provided
- **Donor Profile:**
    - Personal Information
    - Donation History
    - Preferred Communication

**Appointment Management:**

- **Appointment Scheduling Page:**
    - Calendar View
    - Drag-and-Drop Scheduling
    - Filters for Volunteer and Mobile Unit Availability
- **Appointment Details Page:**
    - Client Information
    - Volunteer Assigned
    - Mobile Unit Details
    - Service Type and Status